5 A light-dependent resistor (LDR) has a low resistance in high light intensity and a high resistance in the dark.

(a) Sketch a graph of resistance (*y*-axis) against light intensity (*x*-axis) for an LDR.

[2]

(b) Fig. 5.1 shows part of the electric circuit used to turn on a light when it is dark.

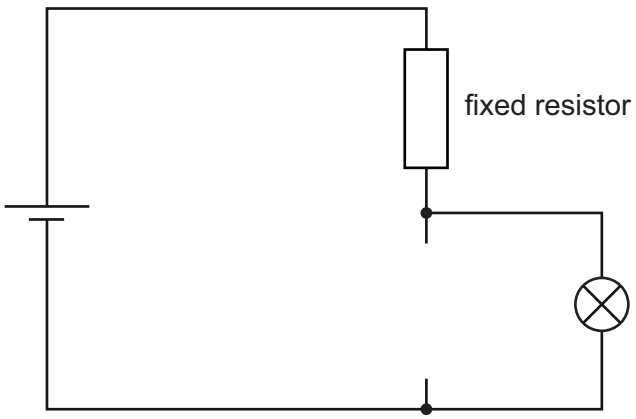

Fig. 5.1

(i) Complete the circuit in Fig. 5.1 with the symbol for a light-dependent resistor (LDR). [1]

(ii) Explain why the lamp is off in the light and the lamp is on in the dark. Use ideas about potential difference (p.d.) in your answer.

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..... [3]

[Total: 6]

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